YOURAN MA

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EDUCATION:

Northwestern University | *McCormick School of Engineering* Sep 2025 – Dec 2026

- *Master of Science* in Computer Science, GPA: 3.95/4.0
- *Relevant Courses*: Software Development Studio, Cryptography, CUDA Programming, Scalable Software Architectures

The Chinese University of Hong Kong, Shenzhen | *School of Data Science* Sep 2021 – Jun 2025

- *Bachelor of Engineering* in Computer Science and Engineering, GPA: 3.6/4.0
- *Dean's List*, School of Data Science, the Chinese University of Hong Kong, Shenzhen
- *Relevant Courses*: Data Structures, Design and Analysis of Algorithms, Computer Architecture, Operating Systems, Machine Learning, Computer Networks, Database Systems, Software Engineering, Parallel Computing

SKILLS:

- **Languages:** Java, C, C++, Python, SQL
- **Frameworks:** Spring Boot, Vue, Qt (C++/QML, Widgets), Rails
- **Databases:** MySQL, SQLite, PostgreSQL. Experienced in database design, query optimization, and deployment
- **Tools:** Git, Linux, Bash, AWS (S3, Lambda, EC2, SQS), Qt Creator, Visual Studio Code

INTERNSHIPS:

Avary.ai **Software Engineer Intern (Part-time)** Nov 2025 – Feb 2026

- Built an AI interviewer product (Avary.ai) in an Agile team, translating customers' requirements into sprint-ready tasks and coordinating PR reviews and merges to deliver backend and frontend features on schedule.
- Improved interview-flow reliability by adding recording duration and silence-detection checks in mock sessions, refining prompts and retry guidance to reduce invalid recordings caused by incorrect user actions.
- Implemented sentence segmentation for the speech system to deliver smoother, low-latency chunked TTS streaming.
- Engineered a scalable recruitment dashboard with Vue 3 frontend and backend APIs for paginated lists, summary cards, and top-ranked views; added filtering and sticky headers to reduce full-data transfer and improve rendering performance.

OSRAM Opto Semiconductors (China) Co., Ltd. **Software Engineer Intern** May 2024 – Aug 2024

- Migrated a legacy PHP-based workflow into a Qt/C++ PDA application supporting barcode scanning, material receiving, and inventory location transfers, improving usability for shop-floor operations.
- Modularized request/response handling by encapsulating Qt's native JSON classes into reusable components, accelerating feature development and improving team productivity.
- Assisted in the Java/Spring Boot backend, adding business logic and database integration for frontend features.
- Worked with production teams to refine workflow details and supported testing, deployment, and release handoff in the company's Alibaba Cloud environment.

PROJECTS:

TFT Smart Hub (Teamfight Tactics Composition Builder) Sep 2025 – Dec 2025

Website: tft-smarthub-517534671b49.herokuapp.com **GitHub:** github.com/NU-CS-Software-Studio-Fall-25/project-tft-smart-hub

- Worked in a Scrum team of 4 people as Tech Lead, coordinating sprint planning, task breakdown, and integration between a Rails RESTful API backend and a Vue 3 + Vite SPA.
- Designed the PostgreSQL data model and Rails MVC endpoints to power scalable team-comp discovery and browsing, adding indexed server-side filtering and pagination to prevent UI freezes as the composition dataset grew.
- Led the Vue 3 + Vite frontend, implementing composition search, editing, and recommendation features, and enhancing website accessibility via semantic markup, ARIA, and keyboard navigation.
- Contributed to CI/CD via GitHub Actions running RSpec (29) and Cucumber (19 scenarios/69 steps), and deployed to Heroku with Postgres, Google OAuth authentication, and SMTP-powered email delivery.

CUDA-Accelerated Attention Forward Pass Dec 2024 – Jan 2025

- Implemented the forward pass of Scaled Dot-Product Attention using PyTorch C++ extensions and CUDA, building naive, tiled, and fused Softmax + P@V kernels with validation and benchmarking scripts.
- Optimized the two major computation stages with shared-memory tiling and warp-level reduction; fused Softmax + P@V to avoid writing probability matrices back to global memory and reduce intermediate memory traffic.
- Built correctness and performance evaluation pipelines in Python; benchmarked inputs with sequence lengths 64–1024 and head dimensions 32–256 on an RTX 2060 SUPER, achieving about 3× speedup over a PyTorch CPU baseline.